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Computer Laboratory Assignment A2 Report

Dimension of the matrix S

The concentration vector, $\vec{c} = (c_1, c_2, c_3, c_4)^T$, represents the concentration of each component involved in the reaction: enzyme (E), substrate (S), complex (C), and product (P), respectively. The rate of change of concentrations can be written as an ODE-system in matrix-vector form:

$$\frac{d\vec{c}}{dt} = \frac{d}{dt}\begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{pmatrix} = \begin{pmatrix} -1 & 1 & 1 & -1 \\ -1 & 1 & 0 & 0 \\ 1 & -1 & -1 & 1 \\ 0 & 0 & 1 & -1 \end{pmatrix}\begin{pmatrix} k_1 c_1 c_2 \\ k_2 c_3 \\ k_3 c_3 \\ k_4 c_1 c_4 \end{pmatrix} = S \cdot \vec{r}(c) \quad (1)$$

, where S is the 4x4 stoichiometric matrix and $\vec{r}(c)$ is a 4x1 column vector of rate expressions.

$$S = \begin{pmatrix} -1 & 1 & 1 & -1 \\ -1 & 1 & 0 & 0 \\ 1 & -1 & -1 & 1 \\ 0 & 0 & 1 & -1 \end{pmatrix} \quad (2)$$

$$\vec{r}(c) = (k_1 c_1 c_2, k_2 c_3, k_3 c_3, k_4 c_1 c_4)^T \quad (3)$$

The initial values $\vec{c}(0)$ are given by the following column vector:

$$\vec{c}(0) = \big(c_1(0), c_2(0), c_3(0), c_4(0)\big)^T = (c_{10}, c_{20}, 0, 0)^T = \vec{c}_0 \quad (4)$$

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Linearity and stiffness

The ODE-system (1) violates the criteria for linearity because the principle of superposition is not satisfied in the vector $\vec{r}(c)$, which contains polynomial operations on the components in $\vec{c}$, specifically the products c_1c_2 and c_1c_4. Therefore, the system is nonlinear. Furthermore, the system (1) is not only non-linear but also stiff. By examining $\vec{r}(c)$, it can be observed that the solution to the system exhibits both rapid and slow changes due to the vector's non-linearity and magnitude of the rate constants, k_i $(i = 1, 2, 3, 4)$. As a result, adaptive time steps will be necessary to effectively solve the equations.

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Solutions for the problem using ode23 and ode23s

Figure 1 illustrates the time evolution (changes over time) of concentrations c_1, c_2, c_3, and c_4. In the four plots, c_1 and c_3 remain relatively constant, while c_2 decreases and c_4 increases. The results indicate similarity between the solvers ode23 and ode23s, as their numerical solutions show minimal differences. Upon closer examination, the rate constant $K1$ appears to impact the results, as c_1 exhibits a slight decrease.

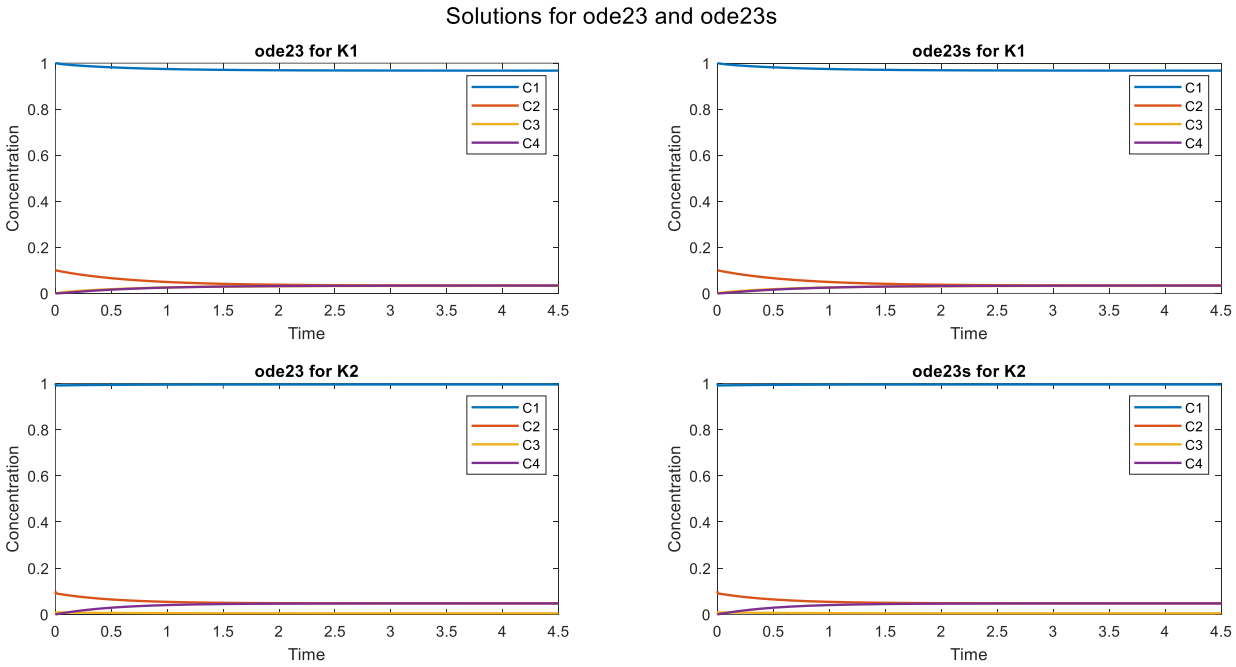


Figure 1: Numerical solutions of the ODE-system for the rate constants K1 and K2, using MATLAB solvers ode23 and ode23s.

1. Comparison between the MATLAB solvers

The performance of the solvers ode23 and ode23s was evaluated for two different rate constants ($K1$ and $K2$) based on the number of steps and execution time. The result is presented in Table 1.

For both cases, ode23 required significantly more time steps than ode23s, particularly for $K2$, where there was a big difference. This indicates that ode23s is more efficient at handling stiff systems. For $K1$, even though ode23 required a greater number of steps compared to ode23s, the execution times remained similar. In contrast, for $K2$, ode23 required almost 100 times more time steps than ode23s, yet its execution time was still slightly lower. Although ode23 exhibited shorter execution time for K2, despite the system's stiffness, the excessive number of time steps suggests that ode23 struggles to handle stiff stiffness efficiently.

In conclusion, given the similar execution times but vastly different time step requirements, ode23s is the more suitable solver for the stiff system with $K2$, while either solver can be used for $K1$.

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Table 1. Comparison of the solvers ode23 and ode23s in terms of the number of steps and execution time for different rate constants (K1 and K2).

Rate Constant	Solver	Number of Time Steps	Execution Time [s]
$K1$	ode23	42	0.058187
$K1$	ode23s	24	0.057226
$K2$	ode23	2004	0.025347
$K2$	ode23s	35	0.034294

2. Linear relations between the concentrations

Figure 2 demonstrates the accuracy of the numerical solvers in maintaining mass conservation. The values remaining close to zero indicate that both solvers effectively conserve mass within the system. Any deviations observed in the plots may be attributed to instabilities arising from the system's stiffness.

As shown in the plots, both $K1$ and $K2$ influence the stiffness of the system. Furthermore, the significant variations of the values in $K2$, compared to $K1$, contribute to increased stiffness. For $K1$, both solvers perform similarly, though ode23 produces smaller residuals, on the order of approximately 10^{-16}, compared to ode23s.

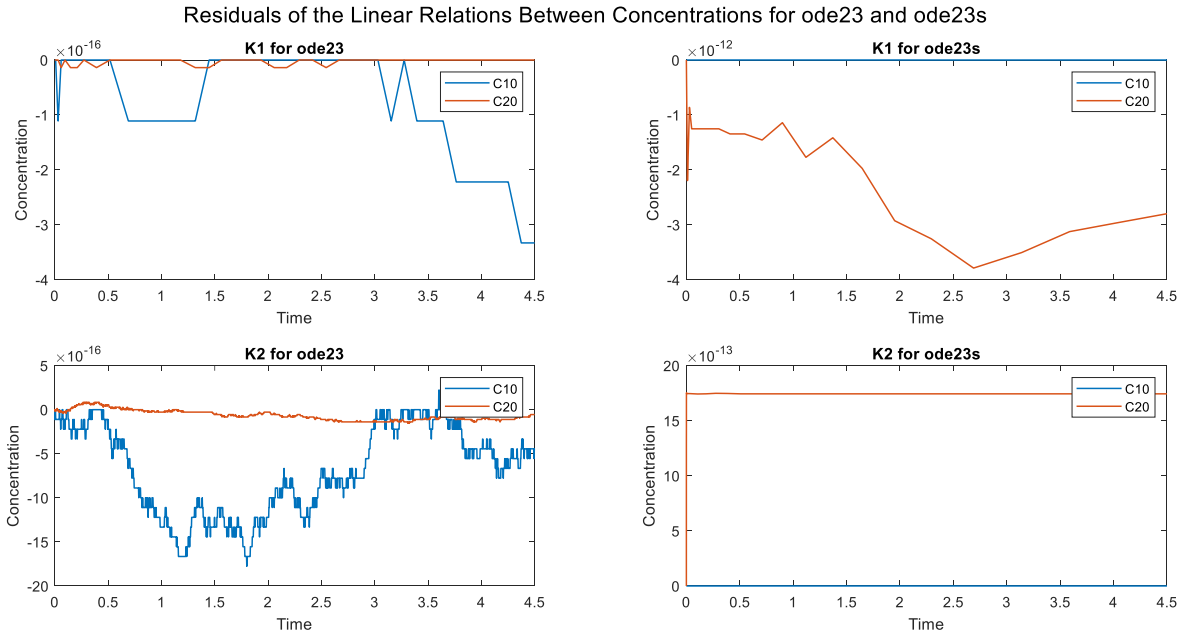


Figure 2: Plots of the linear relations between the concentrations C10 and C20

3. Mass Conservation and Numerical Accuracy in ODE Solutions

The four concentrations in $\vec{c}$ are related through mass conservation, leading to the following linear relationship:

$$c_1 + c_3 = c_{10} \quad (5)$$

$$c_2 + c_3 + c_4 = c_{20} \quad (6)$$

Every measured concentration (c_1, c_2, c_3, c_4) carries some inherent uncertainty due to the numerical approximation by the differential equation solvers, respectively. Since both ode23 and ode2 are one-step solvers, each step relies solely on information from the preceding step. For each step, a (small) local error is included in the approximated value. Consequently, the accuracy of the solution is influenced by the step size and local error, leading to uncertainties in the computed concentrations. Since (6) includes an additional concentration term compared to (5), the numerical calculations based on (6) are likely to introduce larger numerical errors due to the accumulation of uncertainties from the multiple concentration values. Since the system (1) exhibits stiffness, these errors can accumulate over time, leading to deviations from perfect mass conservation. The solver's step-size control mechanism plays a crucial role in mitigating such errors.

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MATLAB code

```
clc, clear, close all
% parameters
K1 = [1 1 10 10];
K2 = [100 1000 10 1];
c0 = [1 0.1 0 0]';
C = zeros(1, 4);

tspan = [0 4.5];

S = [-1 1 1 -1;
    -1 1 0 0;
     1 -1 -1 1;
     0 0 1 -1];

% eigenvalues and condition number to check stiffness
eig(S)
cond(S)

% motivate for the use of options
opts = odeset('RelTol', 1.e-3, 'abstol', 1.e-5);

tic;
[T_k1_23, C_k1_23] = ode23(@odefun1, tspan, c0, opts);
time_k1_23 =  toc;

tic;
[T_k1_23s, C_k1_23s] = ode23s(@odefun1, tspan, c0, opts);
time_k1_23s = toc;

tic;
[T_k2_23, C_k2_23] = ode23(@odefun2, tspan, c0, opts);
time_k2_23 = toc;

tic;
[T_k2_23s, C_k2_23s] = ode23s(@odefun2, tspan, c0, opts);
time_k2_23s = toc;

figure;
subplot(2,2,1)
plot(T_k1_23, C_k1_23, "LineWidth", 1.5)
title('ode23 for K1'); xlabel('Time'); ylabel('Concentration');
legend('C1', 'C2', 'C3', 'C4');

subplot(2,2,2)
plot(T_k1_23s, C_k1_23s, "LineWidth", 1.5)
```

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```
title('ode23s for K1'); xlabel('Time'); ylabel('Concentration');
legend('C1', 'C2', 'C3', 'C4');

%we can see that ode23s takes many fewer steps than ode23
%   which is a stiff problem solver. ode 23 is an explicit solver
subplot(2,2,3)
plot(T_k2_23, C_k2_23, "LineWidth", 1.5)
title('ode23 for K2'); xlabel('Time'); ylabel('Concentration');
legend('C1', 'C2', 'C3', 'C4')

subplot(2,2,4)
plot(T_k2_23s, C_k2_23s, "LineWidth", 1.5)
title('ode23s for K2'); xlabel('Time'); ylabel('Concentration');
legend('C1', 'C2', 'C3', 'C4')

sgtitle('Solutions for ode23 and ode23s');


% residuals and plots
residuals_k1_23_1= C_k1_23(:, 1) + C_k1_23(:, 3) - c0(1);
residuals_k1_23_2 = C_k1_23(:, 2) + C_k1_23(:, 3) + C_k1_23(:, 4) - c0(2);

residuals_k1_23s_1= C_k1_23s(:, 1) + C_k1_23s(:, 3) - c0(1);
residuals_k1_23s_2 = C_k1_23s(:, 2) + C_k1_23s(:, 3) + C_k1_23s(:, 4) - c0(2);

residuals_k2_23_1= C_k2_23(:, 1) + C_k2_23(:, 3) - c0(1);
residuals_k2_23_2 = C_k2_23(:, 2) + C_k2_23(:, 3) + C_k2_23(:, 4) - c0(2);

residuals_k2_23s_1= C_k2_23s(:, 1) + C_k2_23s(:, 3) - c0(1);
residuals_k2_23s_2 = C_k2_23s(:, 2) + C_k2_23s(:, 3) + C_k2_23s(:, 4) - c0(2);

figure;
subplot(2,2,1);
plot(T_k1_23, residuals_k1_23_1, T_k1_23, residuals_k1_23_2, "LineWidth", 1)
title('K1 for ode23'); xlabel('Time'); ylabel('Concentration'); legend('C10', 'C20')

subplot(2,2,2);
plot(T_k1_23s, residuals_k1_23s_1, T_k1_23s, residuals_k1_23s_2, "LineWidth", 1)
title('K1 for ode23s'); xlabel('Time'); ylabel('Concentration'); legend('C10', 'C20')

subplot(2,2,3);
plot(T_k2_23, residuals_k2_23_1, T_k2_23, residuals_k2_23_2, "LineWidth", 1)
title('K2 for ode23'); xlabel('Time'); ylabel('Concentration'); legend('C10', 'C20')

subplot(2,2,4);
plot(T_k2_23s, residuals_k2_23s_1, T_k2_23s, residuals_k2_23s_2, "LineWidth", 1)
title('K2 for ode23s'); xlabel('Time'); ylabel('Concentration'); legend('C10', 'C20')
```

```
sgtitle('Residuals of the Linear Relations Between Concentrations for ode23 and ode23s');

% number of steps
fprintf('Number of time steps K1:\n');
fprintf('ode23: %d steps\n', length(T_k1_23));
fprintf('ode23s: %d steps\n', length(T_k1_23s));

fprintf('Number of time steps K2:\n');
fprintf('ode23: %d steps\n', length(T_k2_23));
fprintf('ode23s: %d steps\n', length(T_k2_23s));

% execution time
fprintf('Execution Time K1:\n');
fprintf('ode23: %f seconds\n', time_k1_23);
fprintf('ode23s: %f seconds\n', time_k1_23s);

fprintf('Execution Time K2:\n');
fprintf('ode23: %f seconds\n', time_k2_23);
fprintf('ode23s: %f seconds\n', time_k2_23s);

function dCdt = odefun1(~, C)
    S =[-1 1 1 -1;
        -1 1 0 0;
        1 -1 -1 1;
        0 0 1 -1];
    K1 = [1 1 10 10];

    r = [K1(1)*C(1)*C(2); K1(2)*C(3); K1(3)*C(3); K1(4)*C(1)*C(4)];
    dCdt = S*r;

end

function dCdt = odefun2(~, C)
    S =[-1 1 1 -1;
        -1 1 0 0;
        1 -1 -1 1;
        0 0 1 -1];
    K2 = [100 1000 10 1];

    r = [K2(1)*C(1)*C(2); K2(2)*C(3); K2(3)*C(3); K2(4)*C(1)*C(4)];
    dCdt = S*r;

end
```